Astro Game Development Kits

Game Submission for Integration Test

AstroGDK version 3.2

For Italian VLT (Comma 6b) and Morocco VLT

S/N: SYSDEV-AGDK-003

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Revision History

Revision	Date (Y/M/D)	Comments
1.6.1	2018/9/28	● Independent document from Section 5 of AstroGDK 1.6.1 with Game submission form added.
2.0	2019/2/8	● For new multi-tab Game lobby (Unity3D), different media files are defined.
2.0 r1	2019/2/21	● Add lost media file bouncing.png ● Add examples of conversion of video files ● Add “Line number” in Game submission form. ● Modify: video resolution should be 1280x800 or 800x600. ● Few typo fixed.
2.0 r2	2019/5/2	● Add specification requirement of game logo for NEXT System. (A.3) ● Explain and request that it’s better to making the first frame and the last frame of bouncing.webm, and the bouncing.png all the same look.
2.0r3	2019/5/8	● Clarify more detail about game icon of NEXT System ● Provide .psd file as standard template and sample game icons for reference while making game icons. ● Indicate that VP9-encoded .webm file is not supported. ● Indicate that option"**-auto-alt-ref 0**": is essential for FFmpeg utility to generate alpha channel into .webm file.
2.1	2019/5/19	● Add and define media file movie_low.avi. It is the low resolution version of movie.avi for Astro H61 platform only. Game team must provide it for game.
2.6	2019/9/30	● Clarify that symlink/hyperlink-type files are not allowed put into package. (sect 1.2.2) ● Lack of ‘-‘ in prefix of option ‘-metadata’ of ffmpeg. (sect B.1.3)
2.7	2020/10/1	● (No changed) ● Indicate that the document is also for Morocco VLT.
2.9-3.2	2022/7/15	● (No changed)

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1 Introduction

The documentation describes how a third party game development partner ("partner") prepares and submits its games to Test Division of Astro Corp. Taiwan ("Astro lab"), for system integration test.

About system integration test, the submitted game will be uploaded into Astro game system ("System") and then downloaded to VLT for testing. "Astro lab" will do tests which focus on correctness and stability of well integration with "System" in categories of process flow, game flow, game recovery, RNG handling, outcome, NVRAM access, accounting, event handling, etc. The system integration test does not care or just partly care of display, audio, game play, and regulation, etc., these are responsibilities of "partner", operator, and / or official test laboratory (e.g. GLI lab).

"Partner" should **prepare "Game Submission Form" and "Game Submission Package"** beforehand, and then **submit both to its agency or operator**. And the later **forwards both materials to contact sales of Astro Corp., mailto:hankwang@astrocorp.com.tw?subject=Game Submission Test <game name> <version>**

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1.1 Game Submission Form

The submission form must contain below information and be provided in form of **.PDF file**.

Basic information of game team or company:

- Company / team name and Country;
- Contact person of technology issues: name, e-mail, and optional Skype, IM account, etc.;
- (others)

Game basic information:

- Game name;
- Game version or revision;
- Game program code name and/or code version/revision (optional);
- Min bet, Max bet, Max win;
- Game features (multiple selections):
  [ ] Free game, [ ] Ultra spin, [ ] Bonus game, [ ] Risk/Double up, [ ] Energy game,
  [ ] System jackpot, [ ] Local jackpot, [ ] Roulette, [ ] Poker, [ ] Scratch,
  [ ] Others:_____
- Line number;
- Theoretical payout rate;
- Additional information or comments to provide (optional);

Data prepared by Test division of Astro Corp. ("Parner" need not provide these)

- Content ID.
- AAMS Game ID.
- Package size (excluding files under media/).

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1.2 Game Submission Package

Game submission package is a collection of directories and files, and could be provided in any form of ftp server connection, link to remote storage, single compressed (*.rar, *.zip) file, etc.

1.2.1 File Types

Three categories totally eight types of materials need to be prepared into submission package:

- Game files:
  - **Game files**: Game-related executable binaries, media files, and data.

- Media files for Downloading agent (downloader):
  - **Downloader Game Logo – Image** (Filename: logo.png): Downloading agent (downloader) will display it to indicate the game in period of downloading.

- Media files for Game lobby:
  - **01 Game Icon – Image** (Filename: icon.png): indicate the game in search result set of Search page and Category (non-Search) page, and for player to select.
  - **02 Game Icon – Animation** (Filename: icon.webm): indicate the game in Category (non-Search) pages, and for player to select.
  - **03 Game Bouncing Icon – Animation** (Filename: bouncing.webm): indicate the game in bouncing button area, and for player to select.
  - **05 Game bouncing Icon – Image** (Filename: bouncing.png): indicate the game in bouncing button area, and for player to select.
  - **06 Game Movie – Video clip** (Filename: movie.avi and movie_low.avi): played on the upper monitor for presenting game features and / or for promotion purpose. File movie_low.avi is just a low-resolution version of movie.avi for Astro H61 platform only.

All above media files will be described in detail in Appendix A.

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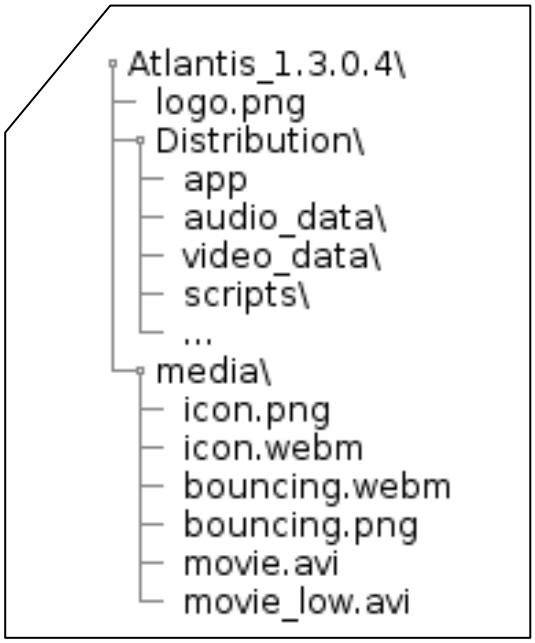
1.2.2 Directory Structure

All above materials must be organized and stored into below directory structure. The whole filled directory structure is just called “Game submission package”:

Directory name	Files Inside	Comment
<Game_name>_<version>	logo.png	Root of game submission package; **logo.png**: Downloader Game Logo – Image;
Distribution	app *.*	**app** will be the file launched by AstroKernel; Game-related run-time files;
media	icon.png	01 Game Icon – Image;
	icon.webm	02 Game Icon – Animation;
	bouncing.webm	03 Game Bouncing Icon – Animation;
	bouncing.png	05 Game Bouncing Icon – Image;
	movie.avi	06 Game Movie – Movie clip;
	movie_low.avi	

Example of a real game submission package (game Atlantis version 1.3.0.4, by Astro Corp.):



Atlantis_1.3.0.4.zip

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Note! All files inside the directory "Distribution\" are hashed for software integrity checking purpose so cannot be changed after official submission (e.g. GLI lab). If requiring changing later, must send a new game submission package. But those files like logo.png and media are not included in integrity check and therefore changeable later anytime.*

Notice! Notice! Don't include libak2api.so in the Game submission package.
Game software must always dynamically link to the libak2api.so of AstroKernel at runtime.

Notice! Notice! Do NOT contain any non-regular types of files in Game submission package.
For example, symlink, hyperlink, device files are all disallowed.

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Appendix A – Specification of Media Files

A.1 Specification of Media File for Downloader

A.1.1 Introduction

In the downloading phase, the downloader will display the game icon and with its downloading progress below in the phase.

- List of file:

Item	File Name	Remark
Downloader Game Logo – Image	logo.png	256x256 PNG still image

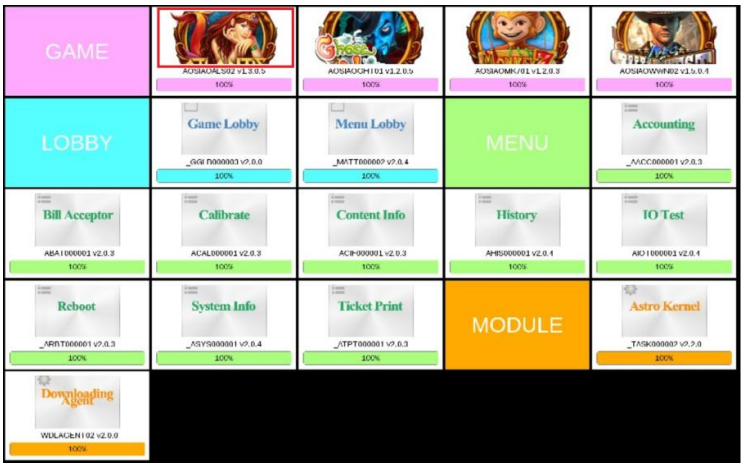

Figure – showing Game logo in downloading phase

A.1.2 Specification of Downloader Game Logo - Image

Item	Specification
File name	logo.png
Format	PNG (Portable Network Graphics)
Dimension	256 x 256 pixels
Pixel format	24 bits or 32 bits, true color with or without ALPHA channel
Transparent	Supported if pixel format contains alpha channel.
Example	logo.png

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A.2 Specification of Media Files for Game Lobby

A.2.1 Introduction

When Game lobby launched, it will load media files from specific directory of individual games installed (refer to Section 1.2.2 above; and Section 4.1.2 of Programming Guide) and show them on the display for indicating the game.

- List of files:

Item	File Name	Remark
01 Game Icon – Image	icon.png	400x280 PNG still image
02 Game Icon – Animation	icon.webm	400x280 WebM (VP8) video, 1 ~ 2 seconds
03 Game Bouncing Icon – Animation	bouncing.webm	400x280 WebM (VP8) video, 2 ~ 4 seconds
05 Game Bouncing Icon – Image	bouncing.png	400x280 PNG still image
06 Game Movie – Video clip	movie.avi	1680x1050 AVI (H.264) video, 1 ~ 2 minutes (for PSM G920 and future HW)
	movie_low.avi	1280x800 or 800x600 AVI (H.264) video 1 ~ 2 minutes (for Astro H61 only)

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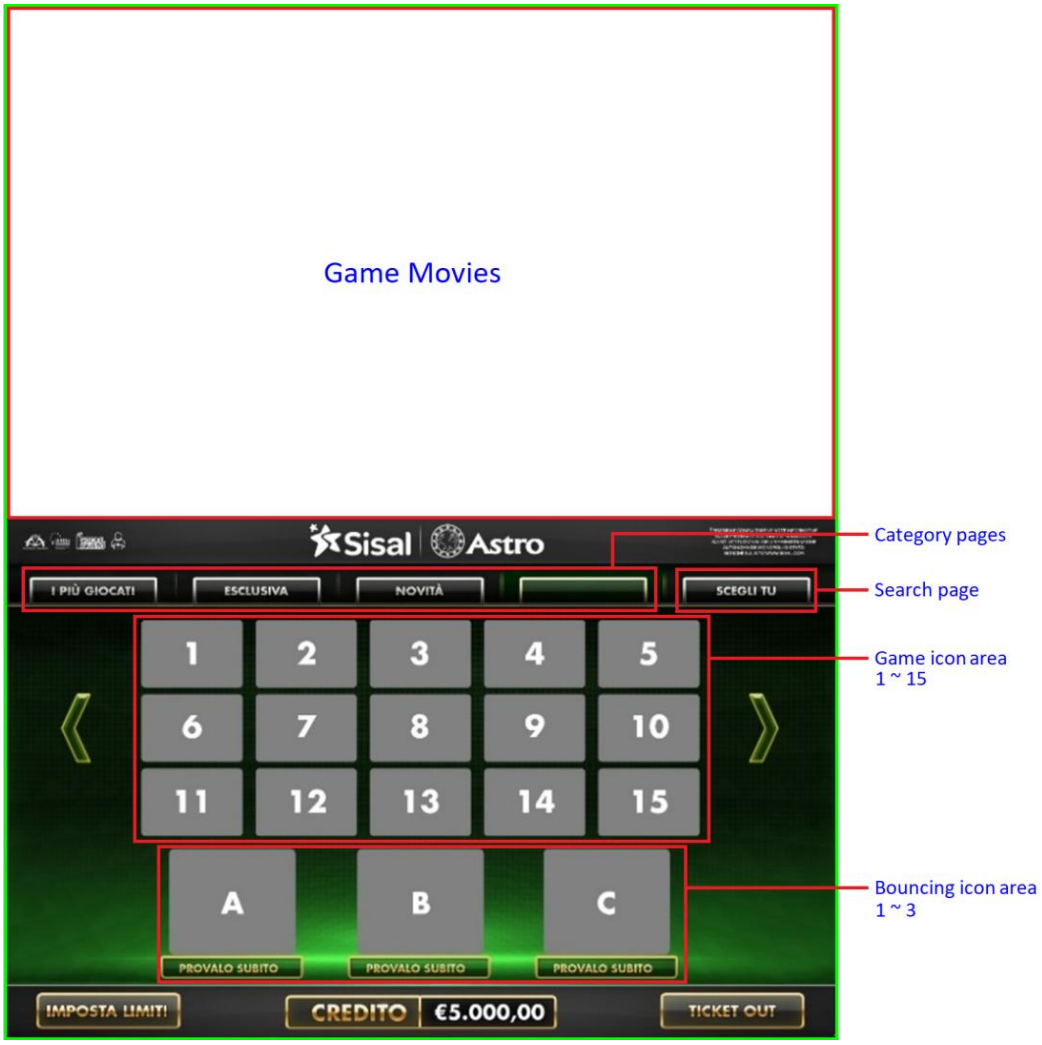
- Basic display layout of Game lobby and Samples



Figures – example of game icons and bouncing icons in Category page

Figure – example of search result set in Search page

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- Display Targets of Media Files

Note! Game package need not provide “04:Bouncing Icon – Label” which are provided by system.

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- Structure Design of Image and Animation

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A.2 Specification in Detail

- 01 Game Icon – Image

Item	Specification
File name	icon.png
Format	PNG (Portable Network Graphics)
Dimension	400 x 280 pixels
Pixel format	24 bits (true color) or 32bits (true color with alpha channel)
Transparency	Supported if pixel format contains alpha channel (ALPHA=0).

- 02 Game Icon – Animation

Item	Specification
File name	icon.webm
File format	WebM
Codec	VP8
Audio	None
Dimension	400 x 280 pixels
Frame rate	30 FPS (frame per second)
Bit rate	3000 ~ 5000 kb/s
Transparency	WebM format supports transparency
Duration	1 ~ 2 seconds

- 03 Game Bouncing Icon – Animation

Item	Specification
File name	bouncing.webm
File format	WebM
Codec	VP8
Audio	None
Dimension	400 x 280 pixels
Frame rate	30 FPS (frame per second)
Bit rate	3000 ~ 5000 kb/s
Transparency	WebM format supports transparency
Duration	2 ~ 4 seconds
Remark	Right after the end of playback of bouncing.webm, bouncing.png will

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	be displayed and keep in the same position. So it's better to make the first frame and the last frame of this bouncing.webm, and the bouncing.png all the same look.

- 05 Game Bouncing Icon – Image

Item	Specification
File name	bouncing.png
File format	PNG (Portable Network Graphics)
Dimension	400 x 280 pixels
Pixel format	24 bits (true color) or 32bits (true color with alpha channel)
Transparency	Supported if pixel format contains alpha channel (ALPHA=0)
Remark	It's better to make this still image just the same look with the first frame and the last frame of bouncing.webm;

- 06 Game Movie – Animation (for PSM G920 and future)

Item	Specification
File name	movie.avi
File format	AVI
Codecs	H.264/MPEG-4 AVC
	MPEG-4 Part 2 (Xvid)
Audio	None
Dimension	1680 x 1050 pixels (1280x800 is also accepted)
Frame rate	30 FPS (frame per second)
Bit rate	4000 ~ 5000 kb/s
Transparency	None
Duration	1 ~ 2 minutes
Remark	Will be randomly displayed on the whole upper monitor on PSM G920 and future platform. For Astro H61, will load and play movie_low.avi instead.

- 06 Game Movie – Animation (for Astro H61 only)

Item	Specification
File name	movie_low.avi
File format	AVI

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Codecs	H.264/MPEG-4 AVC
	MPEG-4 Part 2 (Xvid)
Audio	None
Dimension	1280 x 800 or 800 x 600 pixels
Frame rate	30 FPS (frame per second)
Bit rate	4000 ~ 5000 kb/s
Transparency	None
Duration	1 ~ 2 minutes
Remark	This is just a low-resolution version of movie.avi. movie.avi and movie_low.avi both provide the same look to player. Will be randomly displayed on the whole upper monitor on Astro H61 platform.

Astro system doesn't accept VP9-encoded .WebM file.

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A.3 Specification Requirement of Game Icons for NEXT System

A.3.1 Purpose

All game icons for NEXT system must follow below rules to have required uniform look.

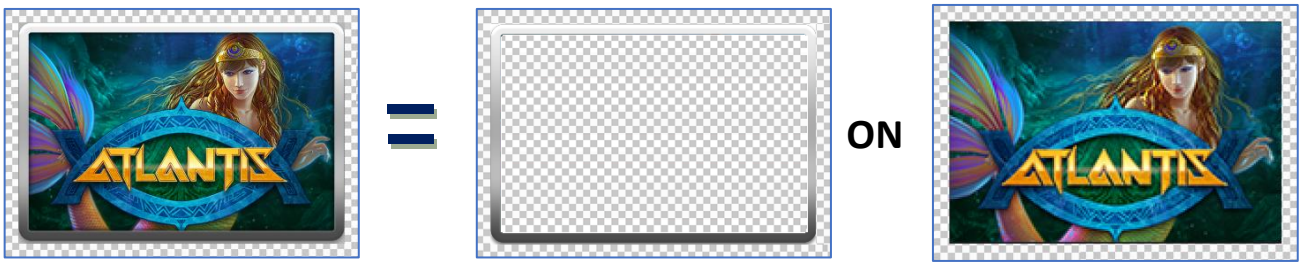
These media files include:

Item	File Name	Remark
01 Game Icon – Image	icon.png	400x280 PNG still image
02 Game Icon – Animation	icon.webm	400x280 WebM (VP8) video, 1 ~ 2 seconds
03 Game Bouncing Icon – Animation	bouncing.webm	400x280 WebM (VP8) video, 2 ~ 4 seconds
05 Game Bouncing Icon – Image	bouncing.png	400x280 PNG still image

You can find the sample .psd file “GameIcon_Image_File_Spec.psd” in directory AstroGDK/doc. It’s the best idea to start making your game icons from ite to definitely make your game icons NEXT system compliant.

A.3.2 Specification

All these media files must support alpha channel (transparency), and always with a border in SILVER (NOT GOLD) color around as below:



Game Icon

Border layer and Image layer in “GameIcon_Image_File_Spec.psd”.

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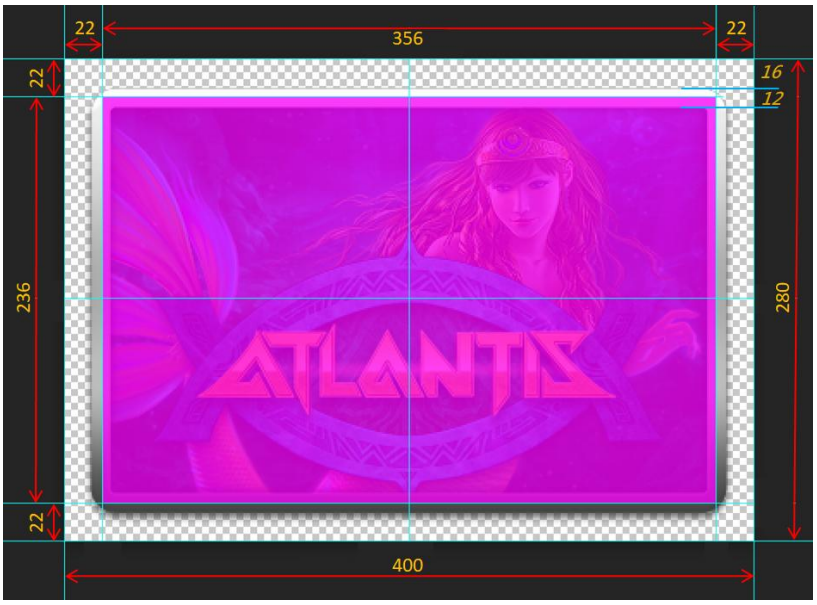
Dimension specification



Dimension Information

Whole image width:	**400**
Whole image height:	**280**
Inner image width:	**356**
Inner image height:	**236**
Border frame width:	***12~***
Border outer side to image side:	***16~***

Units: pixels

(value)~: means it's a rough value.

Note

- Border is drawn upon image. So border width viewed in the game icon is about 12 pixels.
- The border width, the value 12 pixels, is just a rough value for reference only.
  Actually, the border is actually not fixed in its width maybe containing expended shining effect and shadow effect.
- Game team provides the game icons with border and image/animation combined already.
  Game lobby just displays it without any additional border on.

It's the best idea to always start making game icons from the .psd file "GameIcon_Image_File_Spec.psd" for Next system.

Stuff of Appendex A is the revised version of document: VLT_Sisal_Unity_file_spec_0102_2019.pdf

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Appendix B – Examples of Making Media Files

B.1 How to Make WebM (VP8) Video File

B.1.1 Video File Format Conversion

Two options:

- FFmpeg
  Official web: https://www.ffmpeg.org/
  Download: https://ffmpeg.zeranoe.com/builds/
  VP8 encoding guide: https://trac.ffmpeg.org/wiki/Encode/VP8
- Shana Encoder:
  Official web: https://shana.pe.kr/shanaencoder_summary

B.1.2 Usage Example 1 – FFmpeg

Example of using FFmpeg to convert AVI video file to webm file:

(1) Put the "ffmpeg.exe" file in the video source folder;
(2) Open the Command Prompt (cmd) from video source folder;
(3) In the Command Prompt, type below command with [Enter]:

```
ffmpeg -i VideoSourceName.avi -c:v libvpx -crf 10 -b:v 3M -auto-alt-ref 0 -r 30
output.webm
```

Hints:

- "`-crf`": CRF value can be from 4–63, and 10 may be a good starting point. Lower value means better quality;
- "3M": 3 Megabit/s, video bit-rate value (bits per second), choose a higher bit rate if you want better quality, you can choose different values for the bit-rate other than 3M, prefer value 3M ~ 5M (or higher value);
- "`-auto-alt-ref 0`": is essential for generating alpha channel into .webm file.

B.1.3 Usage Example 2 – FFmpeg

Example of using FFmpeg to convert sequence images to webm file:

(1) Put the "ffmpeg.exe" file in the sequence images source folder;
(2) Open the Command Prompt (cmd) from sequence images source folder;

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(3) In the Command Prompt, please type below command with [Enter]:

```
ffmpeg -framerate 30 -i name_%04d.png -c:v libvpx -pix_fmt yuva420p -metadata:s:v:0
alpha_mode="1" -crf 10 -b:v 3M -auto-alt-ref 0 -r 30 output.webm
```

Some examples about filenames:
name_%04d.png means: name_0001.png, name_0002.png, name_0003.png, . . .
name_%05d.png means: name_00001.png, name_00002.png, name_00003.png, . . .

Hints:

- "`-crf`": CRF value can be from 4–63, and 10 may be a good starting point. Lower value means better quality;
- "3M": 3 Megabit/s, video bit-rate value (bits per second), choose a higher bit-rate if you want better quality, you can choose different values for the bit-rate other than 3M, prefer value 3M ~ 5M (or higher value);
- "`-auto-alt-ref 0`": is essential for generating alpha channel into .webm file.

Note! Astro system doesn't accept VP9-encoded .WebM file. So don't use "–c:v libvpx-vp9"

B.1.4 Usage Example 3 – Shana Encoder

Note! Shana Encoder does NOT support converting ALPHA-channel into .WebM format.

Example of using Shana Encoder to convert AVI video file to webm file:

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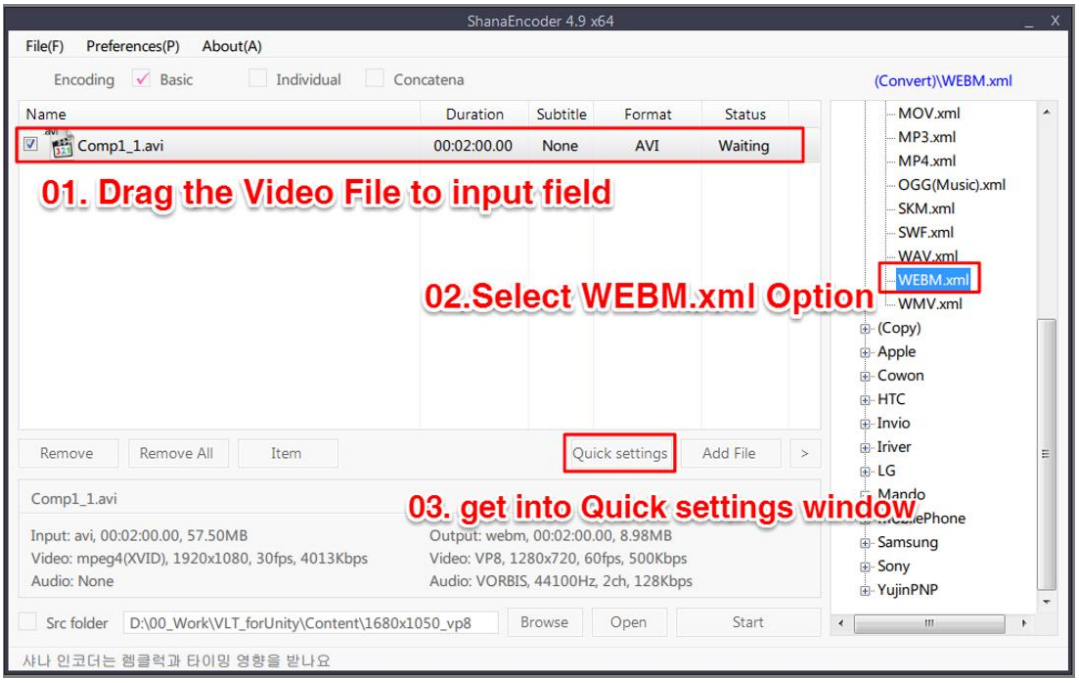
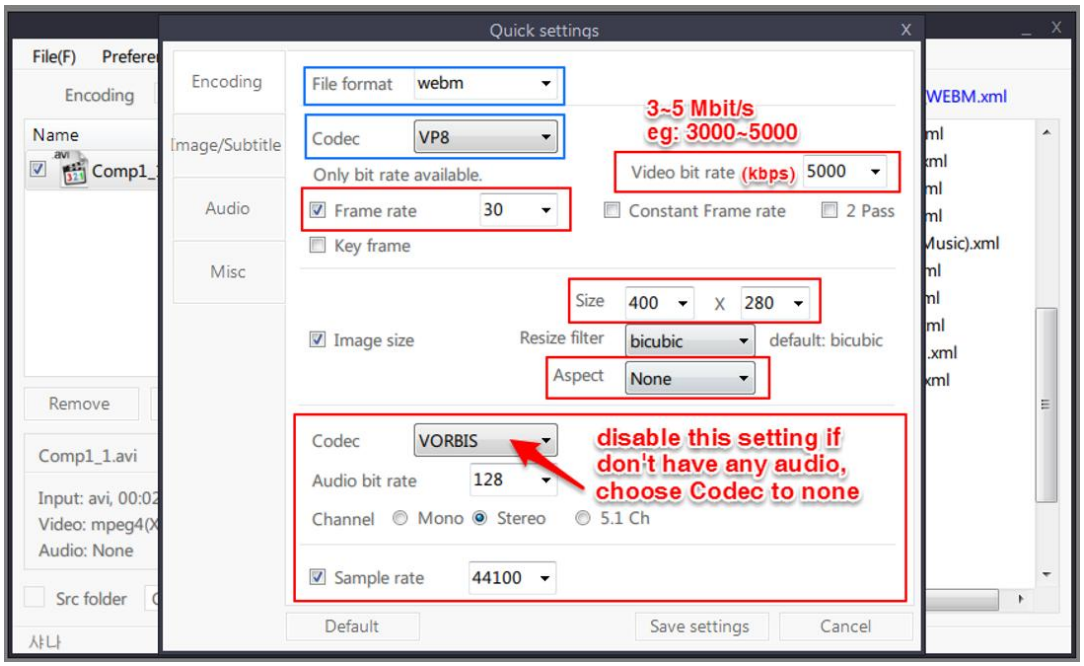
(1) Step 1:



(2) Step 2:



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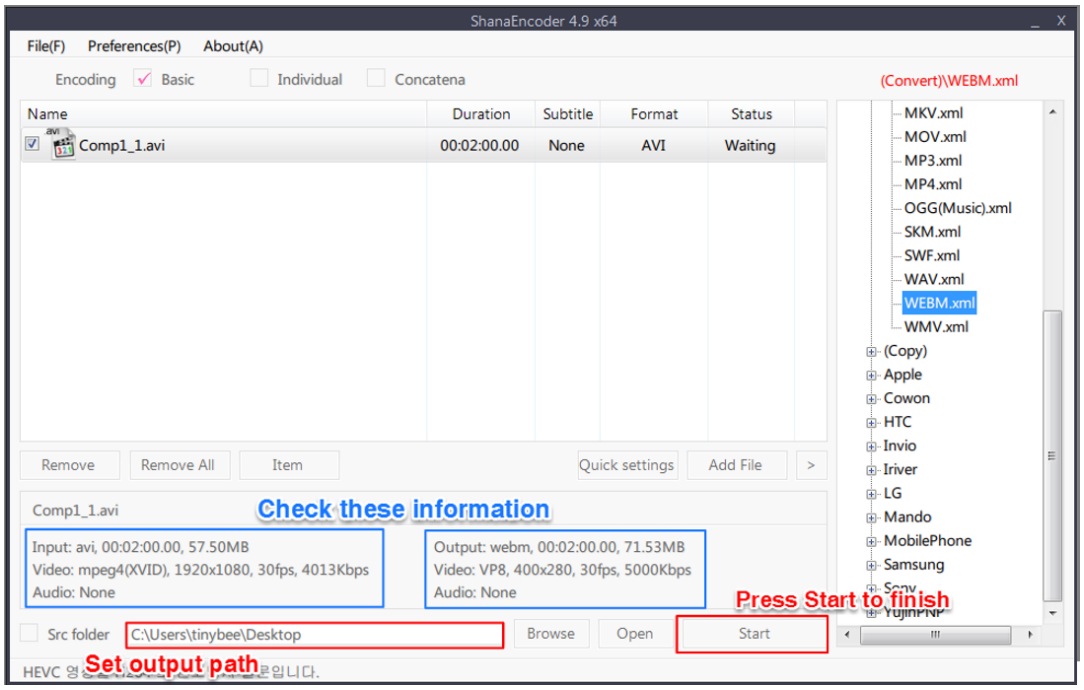
(3) Step 3:



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B.2 How to Make H.264 Video File

Refer to above section, you may occupy FFmeg to convert video file to H.264-format video file just with different parameters.

The H.264 encoding guide: https://trac.ffmpeg.org/wiki/Encode/H.264

Stuff of Appendix B is the revised version of document:
How_to_Make_WebM(vp8)_Video_File_0913_2018.pdf

<End of document>

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